"KENDRIYA VIDYALAYA SANGATHAN, Varanasi Region"

Pre-Board - I Examination 2024-25

Class : X **Max. Marks : 80**

Subject : Mathematics **Max. Time : 3 Hours**

General Instructions:

1. This question paper contains 38 questions. This Question Paper is divided into 5 Sections A, B, C, D and E.
2. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
3. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
4. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
5. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
6. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
7. All Questions are compulsory.
8. However, an internal choice in 2 Questions of section B, 2 Questions of section C and 2 Questions of section D has been provided and internal choice has been provided in all the 2 marks questions of Section E.

	SECTION - A (20 X 1 = 20)	
1	**If n is any natural number , then which of the following numbers ends with digit 0 ?** a) $(3 \times 2)^n$ b) $(5 \times 2)^n$ c) $(6 \times 2)^n$ d) $(4 \times 2)^n$	1
2	**A quadratic polynomial ,the sum of the zeroes is -5 and their product is 6 , is** a) $x^2 +5x+6$ b) $x^2 -5x+6$ c) $x^2 -5x-6$ d) $-x^2 +5x+6$	1
3	**The graph of a pair of linear equation $a_1x + b_1y= c_1$ and $a_2x + b_2y= c_2$ in two variables x and y represents parallel lines** a) $a_1/a_2 \neq b_1/b_2$ b) $a_1/a_2 = b_1/b_2 = c_1/c_2$ c) $a_1/a_2 \neq b_1/b_2 = c_1/c_2$ d) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$	1

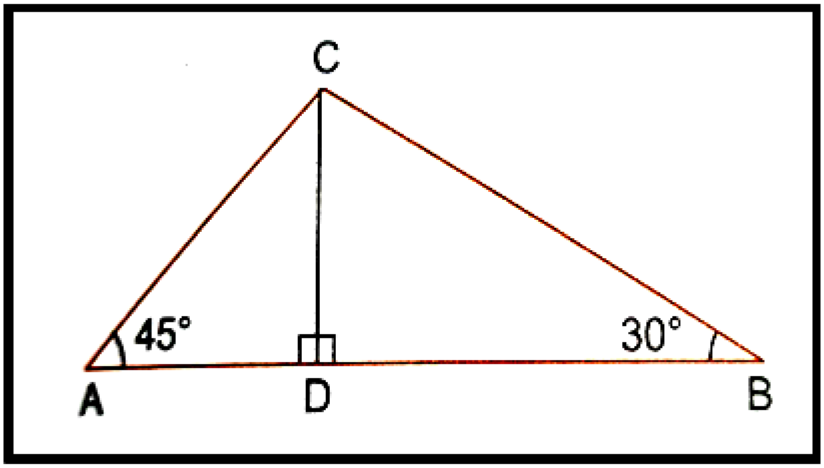
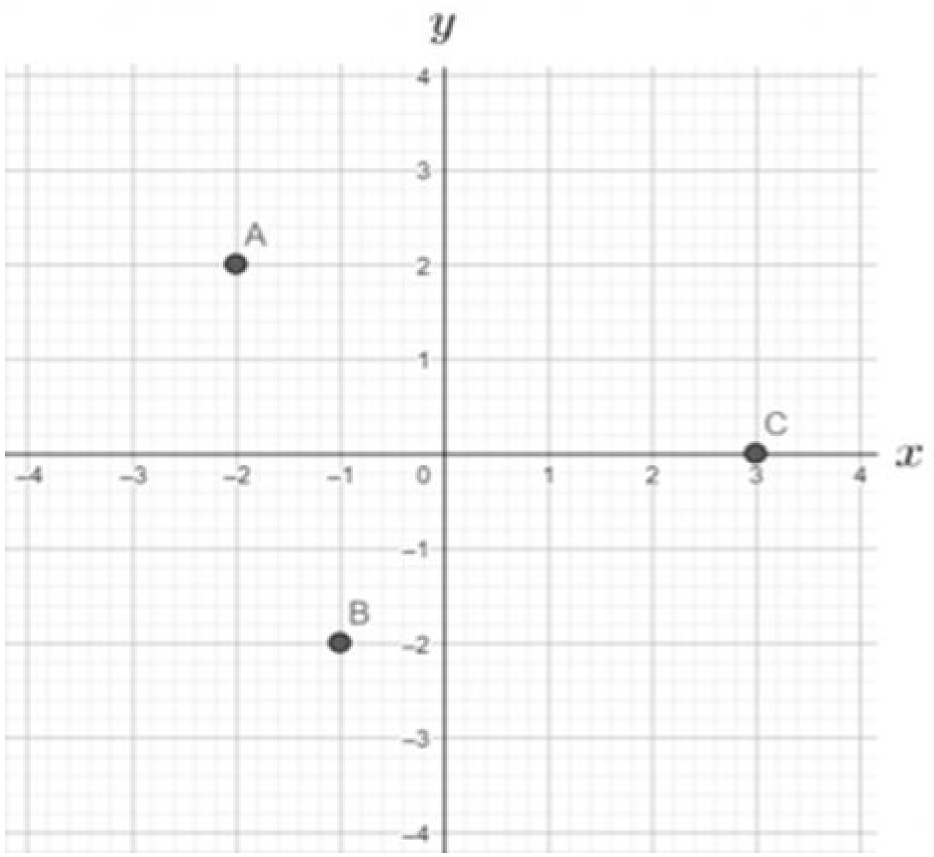
4	Value of k for which x=2 is a solution of the equation $5x^2 - 4x + (2+k) = 0$, is a)10 b) -10 c) 14 d) -14	1
5	The 10th term of the AP: 5, 8, 11, 14, … is (a) 32 (b)35 (c) 38 (d) 185	1
6	The sum of the first five multiples of 3 is: (a) 45 (b) 55 (c) 65 (d) 75	1
7	The distance between the point P(1, 4) and Q(4, 0) is (a) 4 (b) 5 (c) 6 (d) $3\sqrt{3}$	1
8	If the origin is the mid-point of the line segment joined by the points (2,3) and (x,y), then the value of (x,y) is (a) (2, 3) (b) (−2,−3) (c) (−2,3) (d) (2,−3)	1
9	The perimeters of two similar triangles ABC and PQR are 42 cm and 49 cm respectively. AB/PQ is equal to (a) 4/5 (b) 6/7 (c) 7/6 (d) 7/5	1
10	If angle between two radii of a circle is 130^o, the angle between the tangents at the ends of the radii is : (a) 90^o (b) 50^o (c) 70^o (d) 40^o	1
11	If sin A = 4/5 , then value of cos A is: (a)3/5 (b) 1 (c) 2/5 (d)4/5	1
12	If sec θ = 5/4, then evaluate $\tan\theta/(1+\tan^2\theta)$. (a)12/27 (b)12/25 (c)14/25 (d)12/35	1
13	The measure of the angle of elevation of the top of the tower $75\sqrt{3}$ m high from a point at a distance of 75 m from the foot of the tower in a horizontal plane is (a) 60^0 (b) 30^0 (c)45^0 (d)90^0	1
14	If the circumference of a circle increases from 4π to 8π, then its area will become (a) half (b)2 times (c) 4 times (d) none of these	1
15	The surface area of a cube is 216 cm^2 ,its volume is (a) 144 cm^3 (b) 196 cm^3 (c) 212 cm^3 (d) 216 cm^3	1
16	The empirical relationship between mean, median and mode for a distribution is (a)mode = median – 2 mean (b) mode = 3 median – 2 mean (c) mode = 2 median – 3 mean (d)mode = median – mean	1

17	**What is the probability of getting 53 Sundays in 2025?** **(a)1/7 (b)2/7 (c)3/7 (d)4/7**	1
18	**A die is thrown once. What is the probability of getting a number less than or equal to 3?** **(a)1/2 (b)1/6 (c)5/6 (d) none of these**	1
	DIRECTION: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). **Choose the correct option** **(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)** **(b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)** **(c) Assertion (A) is true but reason (R) is false.** **(d) Assertion (A) is false but reason (R) is true.**	
19	**Assertion (A):The length of the tangent drawn fram a point 5 cm away from the centre of circle of radius 3 cm is 4 cm.** **Reason (R): If the angle between two radii of a circle is 130^0 , then the angle between the tangents at the end points of radii at their point of intersection is 50^0 .**	1
20	**Assertion (A): The circumference of a circle is proportional to its diameter.** **Reason (R): The formula C = πd represents the relationship between circumference (C) and diameter (d).**	1
	SECTION - B(5 X 2 = 10)	
21	**Find the HCF of 84 and 144 by prime factorization method.**	2
22	Given a ΔABC and DE ∥ **BC. If AD= 1.5cm, DB=4.5cm, AE =2 cm then find EC.** **OR** **In ΔABC *and* ΔPQR, $\angle A= 60°$. $\angle B = 30°$,$\angle Q =30°$,$\angle R = 90°$.Are the triangles similar? Why?**	2
23	**The length of the minute hand of a clock is 7 cm. Find the area swept by the minute hand in 5 minutes.** **OR** **The length of the minute hand of a clock is 7 cm. Find the length of the arc by the minute hand in 5 minutes.**	2

<table>
<tr><td>24</td><td>Prove that the tangents drawn at the ends of a diameter of a circle are parallel to each other.</td><td>2</td></tr>
<tr><td>25</td><td>If sin A =cos A, find the value of $2\tan^2 A + 1$</td><td>2</td></tr>
<tr><td></td><td>SECTION - C(6 X 3 = 18)</td><td></td></tr>
<tr><td>26</td><td>Prove that $\sqrt{5}$ is an irrational number.</td><td>3</td></tr>
<tr><td>27</td><td>Find the zeroes of the polynomial x^2-5x +6 and verify the relationship between the zeroes and the coefficients.</td><td>3</td></tr>
<tr><td>28</td><td>Find the solution of the pair of linear equations
37x+43y=123, 43x+37y=117?</td><td>3</td></tr>
<tr><td>29</td><td>Prove that the parallelogram circumscribing a circle is a rhombus.
OR
Prove that the lengths of tangents drawn from an external point to a circle are equal.</td><td>3</td></tr>
<tr><td>30</td><td>Prove that: - $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$
OR
If tan (A+B) =$\sqrt{3}$ and tan (A-B) = $1/\sqrt{3}$ and $0^0 < A+B < 90^0$, A>B, Find A and B.</td><td>3</td></tr>
<tr><td>31</td><td>In a test , the marks obtained by 100 students (out of 50) are given below:
<table>
<tr><td>Marks obtained :</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-</td></tr>
<tr><td>Number of students:</td><td>10</td><td>25</td><td>34</td><td>25</td><td>6</td></tr>
</table>
Find the mean marks of the students</td><td>3</td></tr>
<tr><td></td><td>SECTION - D(4 X 5 = 20)</td><td></td></tr>
<tr><td>32</td><td>A plane left 30 minutes late than its scheduled time and in order to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed.
OR
A train, traveling at a uniform speed for 360 km, would have taken 48 minutes less to travel the same distance if its speed were 5km/h more. Find the original speed of the train.</td><td>5</td></tr>
</table>

33	**State and prove Basic Proportionality (Thales) Theorem.**	**5**
34	**A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. Determine the volume of the toy. If a right circular cylinder circumscribes the toy, find the volumes of the cylinder .** **(Take $\pi = 3.14$)** **OR** **The cost of painting the total outside surface of a closed cylindrical oil tank at 60 paise sq. m is Rs 237.60 and the height of the tank is 6 times the radius of the base of the tank.** **(Take $\pi = 22/7$)** **(i) Find the height of the tank.** **(ii) Also find the radius of the tank.**	**5**
35	**Following distribution gives the marks scored by a class of 200 students. If the median of given data is 32, find x and y.** Marks: 0-10, 10-20, 20-30, 30-40, 40-50, 50 Number of students: 20, x, 50, 60, 32, y	**5**
	SECTION - E(3 X 4 = 12)	
36	**Your elder brother wants to buy a car and plans to take a loan from a bank for his car. He repays his total loan of Rs 1,18,000 by paying every month starting with the first installment of Rs 1000. If he increases the installment by Rs 100 every month, answer the following:** **(i) If total installments are 40 then what is the amount paid in the last installment?** **(ii) Find the ratio of the 1st installment to the last installment.** **(iii)What is the amount paid by him in all 30 installments?** **OR** **(iii) What is the amount paid by him in all 20 installments?**	**1+** **1+** **2**
37	**Two boats A and B are approaching a lighthouse CD in mid sea from opposite directions. The angles of elevation of the top of the lighthouse from boats A and B are 45^0 and 30^0 respectively.** **(i) if length of AD is 90 m, then find length of CD.** **(ii) if the length of CD is 60 m then find the length of DB.** **(iii) if the distance between the two boats is 100 m, then find the height of the lighthouse.** **OR** **(iii)if the distance between the two boats is 50 m, then find the height of the lighthouse.**	**1+** **1+** **2**

Marks	0-10	10-20	20-30	30-40	40-50	50
Number of students	20	x	50	60	32	y

	 	
38	**Three friends are standing at point A, B and C. Look at the figure below:** **(i) Find the distance between A and B.** **(ii) Find the distance between B and C .** **(iii) Find the distance between A and C. Which type of triangle is formed by A,B and C .** **OR** **(iii) Find the perimeter of triangle ABC.**  	 **1+** **1+** **2**